CLASS 9 — MATHS · FORMULA SHEET

Chapter 12: **Quadrilaterals** | Quick revision, one page

Angle sum

- Sum of the angles of a quadrilateral $= 360^\circ$ (a diagonal splits it into two triangles).

Parallelogram

- Opposite sides are parallel and equal; opposite angles are equal; adjacent angles are supplementary (180°).
- Diagonals bisect each other (but are not equal in general).
- A quadrilateral with one pair of opposite sides equal and parallel is a parallelogram; so is one whose diagonals bisect each other.
- Area $=$ base $\times$ height; perimeter $= 2(a + b)$.

Special parallelograms

- Rectangle: all angles 90°; diagonals equal and bisect each other.
- Rhombus: all sides equal; diagonals bisect each other at right angles; side $= \sqrt{\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2}$; area $= \frac{1}{2}d_1d_2$.
- Square: all sides equal, all angles 90°, diagonals equal and perpendicular, bisecting each other.
- Rectangle diagonal $= \sqrt{l^2 + b^2}$; square diagonal $= a\sqrt{2}$.

Midpoint theorem

- In $\triangle ABC$, if D and E are the midpoints of AB and AC, then $DE \parallel BC$ and $DE = \frac{1}{2}BC$.
- Converse: a line through the midpoint of one side parallel to the second side bisects the third side.
- The triangle cut off by the midpoint segment has half the perimeter of the original triangle.

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